

Wall-Diabatic Spectral Flow for Monitored-Circuit Endpoint States: A Preregistered Width and Shell Study

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Class-A fermionic adaptive-circuit project

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We specify a flux-threading diagnostic for pure Gaussian endpoint states of a monitored class-A circuit. Earlier static calculations produced almost integer left–right charge transfer only when a fixed-rank occupied subspace was continued through its wall-state crossing. A global maximum-overlap rule does not define that continuation robustly after translation symmetry is lost: a finite-size avoided gap can make it follow the energy-ordered closure instead of the topological spectral-flow branch. The present method isolates the occupied–empty wall doublet, transports the full doublet non-Abelianly, and fixes its occupied line by wall character. It is diabatic with respect to the small internal wall splitting but adiabatic with respect to the external edge–bulk gap. Raw clockwise and counterclockwise subsystem charges are primary. Endpoint quantization is measured, not imposed.

I. WHAT PREVIOUSLY QUANTIZED

The unambiguous positive result is the exact flattened-Hamiltonian benchmark. For a translation-invariant topological slab, it diagonalized the one-particle Hamiltonian separately in each conserved transverse-momentum block. Eigenvectors were assigned between adjacent flux points by maximum overlap, while occupation labels fixed at the start were carried through the wall crossing. It did not refill instantaneous lowest-energy levels after every twist. The continued occupied projector obeyed

$$\Delta N_L \simeq -\sigma, \quad \Delta N_R \simeq +\sigma, \quad q_x = \frac{\Delta N_R - \Delta N_L}{2} \simeq \sigma, \quad (1)$$

where $\sigma = +1$ is counterclockwise (CCW) and $\sigma = -1$ clockwise (CW).

The archived exact data give $\min |q_x| = 0.98081$ over hard and soft walls and both directions on 129- and 257-point meshes. Endpoints agree between those meshes to stored precision, the largest charge residual is 1.14×10^{-13} , and explicit reversal returns the projector with normalized Frobenius residual below 5×10^{-18} . These values are revalidated from archived CSV tables before a new launch.

The static monitored-endpoint calculation used the same direct charge observable but selected the entire rank- r subspace globally. At 20×24 , $n_{\text{shell}} = 1$, 71 of 100 soft-wall and 36 of 100 hard-wall samples transported one charge on a 64-interval grid. On 128 intervals the counts became 64 and 30. The charge was not mismeasured; the selected branch changed. Event samples also had smaller occupied–empty crossing gaps than closure samples. Arbitrarily fine global continuation resolves the small finite-size avoided gap and tends to follow the branch that closes.

II. ENDPOINT STATE AND THREADED PARENT

A number-conserving pure Gaussian state has an occupied frame $F_0 \in \mathbb{C}^{N \times r}$,

$$F_0^\dagger F_0 = \mathbf{1}_r, \quad P_0 = F_0 F_0^\dagger. \quad (2)$$

The physical object is P_0 , not the basis chosen for F_0 . Define $H_0 = \mathbf{1}_N - 2P_0$. In the production uniform gauge, transverse flux gives

$$\tilde{H}_{rs}(\phi) = (H_0)_{rs} \exp[i\phi d_y(r, s)/N_y], \quad (3)$$

$$H(\phi) = \frac{1}{2}[\tilde{H}(\phi) + \tilde{H}^\dagger(\phi)], \quad (4)$$

where d_y is the minimum-image transverse displacement. A large-gauge map G relates the endpoints; all endpoint comparisons return to the starting gauge.

For $\sigma = \pm 1$, the mesh is

$$\phi_j^{(\sigma)} = -\sigma 10^{-7} + \sigma \frac{2\pi j}{256}, \quad j = 0, \dots, 256. \quad (5)$$

The signed regulator chooses one side of an ambiguous crossing. It is not a fit parameter and cannot make a non-topological response quantized.

III. THE ORDER-OF-LIMITS PROBLEM

Near $\phi_\star$, an occupied state on one wall and an empty state on the other approach the rank boundary. Finite wall separation hybridizes them, producing an internal splitting δ_{int} . Energy ordering defines a finite-system adiabatic branch which closes after one flux quantum. Topological spectral flow uses a different order of limits: it remains adiabatic to bulk states while crossing the exponentially small inter-wall splitting diabatically.

A global rank- r maximum-overlap selector makes one global yes/no choice at this crossing. Small sample or grid changes can alter it. Grid refinement alone is not a cure because it follows the finite-size avoided crossing more

faithfully. Increasing wall separation reduces δ_{int} , but a robust rule must also distinguish the internal splitting from the external gap to spectator states.

IV. WALL-DIABATIC EDGE TRANSPORT

At each flux, a small spectral block $E(\phi)$ contains the occupied–empty wall crossing. The primary block has dimension two; rank four is a sensitivity test. Let R_L, R_R project onto windows around the physical walls. Within the block, diagonalize

$$B_E = E^\dagger(R_R - R_L)E. \quad (6)$$

Its extreme eigenvectors are maximally left- and right-polarized combinations. Their eigenvalues and combined wall weight are saved. The internal gap δ_{int} is ignored when fixing the diabatic wall label. The external gap

$$\delta_{\text{ext}} = \min_{i \in E, a \notin E} |E_i - E_a| \quad (7)$$

separates the block from spectator states. A path is marked unresolved if the block cannot be isolated or lacks definite wall character. It is reported, not forced into a zero/one class.

The whole edge block is aligned between adjacent fluxes. For orthonormal bases E_j, E_{j+1} , form

$$S_j = E_j^\dagger E_{j+1} = U_j \Sigma_j V_j^\dagger. \quad (8)$$

The polar link aligns $E_{j+1} V_j U_j^\dagger$ with E_j . This is discrete non-Abelian Kato transport: rotations inside the nearly degenerate edge multiplet are gauge freedom. Every link's minimum singular value is saved. Wall-operator eigenvectors fix physical left/right labels inside the transported block.

The initially occupied edge line is carried through this block without being reselected by energy. Occupied spectator modes are continued as an isolated subspace. Their projector plus the occupied edge line gives fixed-rank $P_{\text{cont}}(\phi)$. The prescription is adiabatic from edge block to bulk but diabatic within the finite-size wall doublet. It reproduces the occupation-label convention that conserved momentum made unambiguous in the exact benchmark.

V. DIRECT CHARGE AND ENDPOINT DEFECT

With half-cylinder region projectors,

$$N_L(\phi) = \text{tr}[R_L^{\text{half}} P_{\text{cont}}(\phi)], \quad (9)$$

$$N_R(\phi) = \text{tr}[R_R^{\text{half}} P_{\text{cont}}(\phi)], \quad (10)$$

$$q_x(\phi) = \frac{1}{2} \{ [N_R(\phi) - N_R(\phi_0)] - [N_L(\phi) - N_L(\phi_0)] \}. \quad (11)$$

Raw CCW and CW paths are reported separately for every sample. No direction-odd average is substituted.

At the endpoint define

$$D = G^\dagger P_{\text{cont}}(\phi_0 + \sigma 2\pi) G - P_0. \quad (12)$$

A clean rank-preserving one-particle transfer has one eigenvalue near +1, one near −1, and all others near zero. The positive and negative defect modes must lie on opposite walls. The spectrum of D , its mode densities, and Eq. (11) are independent views of the same transfer.

Robustness to an arbitrary cut is tested by evaluating q_x at several bulk-plateau cuts and recording normalized center-of-charge displacement. Neither replaces direct subsystem charge. Polar links also define an edge holonomy, but a one-parameter Wilson determinant alone is not identified with the pumped integer.

VI. WIDTH AND SHELL CAMPAIGN

No monitored dynamics is rerun. Authoritative endpoint ensembles contain 100 soft-wall and 100 hard-wall states in each of eight cells:

$$(N_x, N_y) = (20, 24), (24, 24), (28, 24), (32, 24), \quad (13)$$

for both $n_{\text{shell}} = 1$ and the dense-shell controller. Each endpoint pair stores both directions, giving 1600 primary atomic NPZ/completion pairs and 3200 primary raw directional paths. When the A100 endpoint campaign is accepted, 25 GPU bridge endpoints per wall are also analyzed for the three cells whose primary endpoints came from the legacy CPU campaign. This adds 150 non-primary pairs without pooling backends, for 1750 base pairs total on the GPU route. The bridge is absent on the all-CPU fallback route. Input pairs, configuration and source hashes, frame dimensions, rank, complex128 dtype, and Gram residuals are verified.

The primary mesh has 256 intervals, wall-window radius two, and block rank two. Development checks repeat selected samples on 128 and 512 intervals, wall window radii two and three, and block ranks two and four. Outputs retain wall-diabatic, ordinary-global-overlap, and instantaneous-refill charge histories; gaps; wall-operator spectra and weights; polar-link conditioning; resolution status; endpoint defect data; multi-cut charge; center displacement; source real-space Chern markers; and numerical residuals.

One endpoint and both directions form one durable result/completion pair. Restart verifies name, byte count, SHA-256, configuration and source identity, and endpoint dependency. Invalid pairs rerun; verified pairs are skipped.

The standard source route is the imported A100 campaign. If its durability benchmark is rejected, a separate CPU fallback generates only the 1000 missing primary endpoints: five samples are aggregated per shard, two 28-worker NUMA lanes occupy cores 0–27 and 28–55, and the canonical CPU engine writes state/RNG checkpoints every five cycles. A shard is published only after

all five endpoints verify. The CPU fallback does not synthesize a GPU bridge; the bridge is meaningful only as an independent backend check.

VII. CONTROLS AND ACCEPTANCE

The archived exact pump is the positive control. Its mesh agreement, orientation reversal, true undo, and large-gauge equivalence must pass before launch. Ordinary global overlap and instantaneous refill are saved inside every task as protocol controls. When matched trivial, Chern-conjugated, and seam-gauge controls are present, preregistered gates are

$$\max |q_x|_{\text{trivial}} < 0.05, \quad (14)$$

$$\max |q_x^{C=+1} + q_x^{C=-1}| < 0.05, \quad (15)$$

$$\max |q_x^{\text{uniform}} - q_x^{\text{seam}}| < 10^{-8}. \quad (16)$$

The executable exact control is the symmetry-resolved legacy calculation, not a twist of the single zero-flux surrogate $1 - 2P_{\text{exact}}(0)$. For every value of ϕ we rebuild the canonical OW functions and hence $H_{\text{exact}}(\phi)$, transform to the 40 conserved k_y blocks, match the complete eigenbasis within each block by maximum overlap, and polar-align degenerate clusters. The initially occupied labels are then retained around the loop. This is precisely the momentum-resolved spectral-flow convention that made the equilibrium wall branch unambiguous. Because the exact wall band contains adjacent momentum sectors, its global rank-boundary pair is not required to satisfy the monitored-state two-level isolation threshold $\Delta_{\text{ext}} \geq 0.1$; translation-sector conservation supplies the stronger label. Seam-gauge vectors are transformed back to the common uniform gauge before overlap comparison.

Numerical validity requires charge residual below 10^{-10} , projector residual below 10^{-10} , finite diagnostics, and a verified atomic pair. Exact controls require $|q_x| > 0.97$ and endpoint mesh changes below 10^{-4} across the 128-, 256-, and 512-interval meshes (the archived 129/257 comparison is substantially tighter). The preregistered endpoint sensitivities are named **M128**, **M512**, **seam.M256**, **radius3.M256**, and **rank4.M256**. Unresolved edge blocks are flagged rather than repaired after inspection.

Endpoint quantization is not an acceptance gate. Scientific outputs are sample-wise CW and CCW q_x distributions and the correctly oriented variable σq_x . We report fractions with $\sigma q_x > 0.5$ and $\sigma q_x > 0.9$, and the fraction in $0.95 \leq \sigma q_x \leq 1.05$, together with resolved/unresolved fractions. The crossing diagnostics are evaluated at the selected minimum-internal-gap index, rather than by combining unrelated global minima. Width scaling fits $\delta_{\text{int}} \propto \exp[-W/\xi]$ against the physical wall separation $W = N_x/2$ and bootstrap-resamples trajectories separately within each width. The optional GPU bridge is compared with the matching CPU strata as an independent, unpaired ensemble, including a GPU-indicator regression with cell-wall-direction fixed effects

and a stratified bootstrap interval; it is never pooled into the primary estimates. Remaining outputs compare shell dependence and agreement among direct charge, multi-cut plateaus, and the ± 1 defect pair.

VIII. REPRODUCIBILITY, DURABILITY, AND PRESENT STATUS

The locked spectral campaign identifier is **N20_24_28_32x24_wall_diabatic_spectral_pump_s100_v1**. Its machine-readable contract is **campaign_config.wall_diabatic_spectral_pump_s100_v1.json**; the local execution profile, source-selection rule, CPU affinity, worker count, and expected pair counts are recorded separately in **local_profile.wall_diabatic_spectral_pump_s100_v1.json**. The human operator sequence and recovery rules are fixed in **WALL_DIABATIC_WIDTH_SWEEP_RUNBOOK.md**. These three artifacts must agree before launch.

The missing endpoint revision is **wall_pump_width_endpoints_s100_v1**, with root seed 2026090701. The self-contained Colab bundle requires a 40-GB A100, stages execution under **/content**, and calls the canonical GPU entry point **classA_U1FGTN_gpu.run_markov_circuit**. Its 1000 missing primary and 150 bridge endpoints are published in five-trajectory shards. One stable rolling checkpoint is written every five circuit cycles and binds the native occupied frames, ranks, partial charge observer, and NumPy and Torch CPU/CUDA random-number states. A completed shard is recognized only from its NPZ and completion JSON after final-path readback, byte-count, configuration, and SHA-256 verification.

The endpoint benchmark first sweeps candidate execution batches 10, 20, 40, 60, 80, 100 on the largest dense soft-wall cell. It requires at least 8 GiB measured A100 headroom and then times all ten missing protocol-width-wall arms with production-shaped five-trajectory, 48-cycle runs. The complete projected A100 workload must not exceed 40 hours, one half of the locked approximately 80-hour 56-core local forecast. At least 25 GiB of output storage is required before production. A failed gate is preserved and selects the canonical two-lane CPU fallback; it is never weakened after observing the timing.

The GPU engine backend is determined by the correction support: finite $n_{\text{shell}} = 1$ uses its local sparse backend, whereas $n_{\text{shell}} = \text{dense}$ (encoded as **None**) uses the dense backend. A first launch on September 7 stopped during construction of the dense benchmark model because the launcher had incorrectly forced the local backend. The corrected mapping is covered for every locked campaign cell. That attempt preceded benchmark dynamics and produced no benchmark receipt, endpoint shard, or scientific datum.

The spectral calculation is local CPU work on physical cores 0–55 with 56 one-thread workers unless a separate benchmark demonstrates both complex128 agreement below 10^{-10} and at least twofold throughput against a

same-host 56-affinity CPU reference. One endpoint, containing both raw CCW and CW paths, is the atomic resume boundary. Its completion binds the exact endpoint dependency, scientific configuration, source hashes, result byte count, and SHA-256. A worker failure leaves that pair pending and writes a diagnostic; it cannot be counted as completion. The bridge uses a disjoint output collection and remains excluded from primary statistics.

At the date of this preregistration, the corrected software, generated notebook, source-copy parity, exact archived-reference check, small-frame CPU/GPU frozen schedule parity, checkpoint restoration, task expansion, control definitions, and analysis interfaces have passed their maintained local tests. The corrected A100 campaign subsequently passed its locked gate with a 30.043-hour projection and execution batch 80. Its repository import contains all 1000 new primary endpoints and all 150 GPU bridge endpoints in 230 verified shards (16,905,617,976 bytes), with zero failed result/completion pairs. This source inventory is fixed by `imported_endpoints/wall_pump_width_endpoints_s100_v1/DOWNLOAD_MANIFEST.json`. An initial exact-control launch was stopped after it exposed

the incorrect zero-flux-surrogate construction. The corrected implementation rebuilds the full exact flux family and reproduced $q_x = (1.000000, -0.99999999)$ in a coarse eight-interval live check, with instantaneous closure at 10^{-8} , true-undo error below 3×10^{-17} , and source Chern marker 0.999958. No production wall-diabatized endpoint result or sensitivity product is claimed until the corresponding atomic outputs verify and the locked analysis is complete.

IX. INTERPRETIVE BOUNDARY

This is a spectral diagnostic of each saved state. It is not a physical-time Schrödinger ramp, online monitored pump, frozen-record replay, or tangent cocycle. Here *adiabatic* does not mean a fast quench; it means ignoring the internal splitting of distant wall modes while keeping their combined subspace isolated from the bulk. The construction can make the intended branch definition reliable. It cannot manufacture a Chern pump if a saved state lacks a resolved topological wall subspace.